Fundamental Cybersecurity Concepts

Core Definition

Cybersecurity is the practice of protecting systems, networks, devices, and sensitive digital data from unauthorized access, disruption, damage, or cyber attacks.

Real-World Threat Scenarios

- **Account Compromise** — Unauthorized access to personal or organizational social media accounts, such as an Instagram account hijacking.
- **Financial & OTP Fraud** — Social engineering is used to trick users into revealing One-Time Passwords (OTPs) or approving fraudulent transactions.
- **Email Exploits** — Deceptive messages may deliver malicious payloads, credential-harvesting pages, or harmful attachments.
- **Malware Attacks** — Malicious software may execute on endpoints and damage systems, steal information, or provide unauthorized access.

Key Cyber Threat Vectors

Phishing

Phishing uses deceptive emails, calls, messages, or websites to impersonate trusted entities. Its main objective is often to steal credentials, financial information, personal data, or make the victim perform an unsafe action.

Forms of Phishing

- **Email phishing** — Fake emails impersonate trusted organizations to steal passwords, financial details, or other information.
- **Spear phishing** — Highly targeted phishing aimed at a particular person or organization, often using personalized information.
- **Whaling** — Spear phishing targeting senior executives or other high-value individuals.
- **Smishing** — Phishing through SMS or text messages, often containing malicious links.
- **Vishing** — Phishing through voice calls in which scammers impersonate banks, government agencies, technical support, or other trusted parties.
- **Clone phishing** — A legitimate-looking message is copied and modified so its link or attachment becomes malicious.
- **Social-media phishing** — Fake accounts, messages, or login pages are used to obtain credentials or personal information.
- **Pharming** — Victims are redirected to a fraudulent website even when they intend to visit a legitimate one.

- **QR phishing (quishing)** — Malicious QR codes direct victims to fake websites designed to capture information.

How to Recognize Phishing

- Check the sender address and domain carefully rather than relying only on the display name.
- Look for urgency, threats, unusual requests, or offers that seem too good to be true.
- Verify the real destination of links before opening them.
- Never disclose passwords, OTPs, recovery codes, or banking credentials because of an unsolicited request.
- Use official websites or known contact numbers to verify suspicious requests.

Malware

Definition

Malware means malicious software. Examples include viruses, ransomware, spyware, and trojans. Malware can damage systems, exfiltrate confidential data, monitor users, or lock files for ransom.

Common Types of Malware

- **Virus** — Attaches itself to legitimate files and spreads when the infected file is executed.
- **Worm** — Can replicate and spread automatically across networks without requiring the user to execute every copy.
- **Trojan Horse** — Pretends to be legitimate software but performs malicious actions after installation.
- **Ransomware** — Encrypts or blocks access to data and demands payment from the victim.
- **Spyware** — Secretly monitors the victim and collects information.
- **Keylogger** — Records keystrokes, potentially capturing usernames, passwords, and messages.
- **Rootkit** — Attempts to hide malicious activity while maintaining privileged access to a system.
- **RAT (Remote Access Trojan)** — Provides an attacker with remote control over an infected machine.
- **Botnet Malware** — Turns infected devices into remotely controlled bots that can be used together for malicious activities.
- **Adware** — Displays unwanted advertisements and may track user activity or redirect the browser.

Basic Malware Prevention

- Keep operating systems, browsers, applications, and security tools updated.
- Install software only from trusted sources and verify unexpected attachments before opening them.
- Use endpoint protection or antivirus and enable appropriate security controls.
- Maintain regular backups, including protected copies that can help against ransomware.

- Use least privilege so normal users do not operate with unnecessary administrator rights.

Brute Force Attacks

Definition

A brute-force attack is an automated, systematic attempt to guess credentials by trying many combinations. Its impact can include cracking passwords, bypassing authentication, and gaining unauthorized access.

Types of Brute-Force Attacks

- **Simple Brute Force** — Tries many possible combinations systematically, such as every possible PIN.
- **Dictionary Attack** — Uses a list of common passwords or words such as password, admin, or qwerty.
- **Hybrid Attack** — Combines dictionary words with modifications, such as password123 or admin2026.
- **Credential Stuffing** — Uses previously leaked username/password combinations against other websites. It works because people sometimes reuse passwords.
- **Password Spraying** — Tries one common password against many accounts rather than many passwords against one account.
- **Reverse Brute Force** — Starts with a known/common password and tries it against many usernames.

How to Defend Against Brute Force

- Use long, unique passwords or passphrases for every account.
- Enable multi-factor authentication (MFA) wherever possible.
- Use rate limiting and login throttling.
- Monitor repeated failed logins and unusual authentication patterns.
- Prevent password reuse and use compromised-password checks where appropriate.

Security Operations Center (SOC)

Definition

A Security Operations Center (SOC) is a centralized command post where a dedicated security team continuously monitors, detects, analyzes, and responds to cybersecurity incidents.

Core SOC Responsibilities

- **System Monitoring** — Continuous visibility across endpoints, servers, applications, and networks.

- **Threat Detection** — Identifying suspicious patterns, anomalies, malicious activity, and unauthorized access attempts.
- **Incident Response** — Swiftly containing, mitigating, investigating, and remediating security events to minimize damage.

Typical SOC Workflow

- 1. Collect — Gather logs and security telemetry from endpoints, servers, network devices, applications, and cloud services.
- 2. Detect — Identify potentially malicious events using rules, analytics, threat intelligence, or behavioral indicators.
- 3. Triage — Determine severity, scope, affected assets, and whether the alert is a true positive or false positive.
- 4. Investigate — Correlate logs and evidence to understand what happened and how the activity occurred.
- 5. Contain — Limit the incident's impact, such as isolating an affected endpoint or disabling a compromised account.
- 6. Eradicate & Recover — Remove the cause of the incident, restore systems safely, and verify that malicious activity has stopped.
- 7. Document & Improve — Record findings, update detections, and improve controls to reduce recurrence.

Common SOC Technologies

- **SIEM** — Collects and correlates security logs and events to help analysts detect suspicious activity.
- **EDR** — Monitors endpoint activity and helps investigate and respond to threats on computers and servers.
- **IDS/IPS** — Detects suspicious network traffic; an IPS can also take preventive action depending on configuration.
- **Threat Intelligence** — Provides information about known malicious indicators, techniques, campaigns, and threats.

Suspicious Link & URL Analysis

How to Identify a Suspicious URL

- **Domain Name & Typosquatting** — Look for subtle misspellings such as micros0ft.com or paypa1.com.
- **Subdomain Hierarchy** — Identify the actual registered domain. For example, in login.bank.com.attacker.com, the controlling domain is attacker.com.
- **Protocol Check** — Verify whether the site uses HTTPS. HTTPS encrypts traffic in transit but does not by itself prove that the website is trustworthy.

- **URL Shorteners** — Be cautious with shortened URLs because they can obscure the final destination.
- **Direct File Extensions** — Be cautious with links that directly point to executable files or archives such as .exe, .iso, .zip, or .scr.

Safe Analysis Checklist

- Do not open a suspicious link simply to see what happens on a personal or production device.
- Check the complete domain and path before deciding whether a link is legitimate.
- If a message claims to be from a bank, company, or service, navigate to the organization's official website independently.
- For security training, analyze suspicious URLs only in an isolated, authorized environment.

Git Basics

Definition

Git is a tool used to track changes in code and help developers collaborate.

Git vs GitHub

Git is the version-control tool. GitHub is an online platform where Git repositories can be hosted and shared.

Basic Git Commands

git init

Initializes a Git repository in the current project folder.

```
git init
```

git status

Shows the current status of the repository, including modified and untracked files.

```
git status
```

git add .

Stages changes from the current directory for the next commit.

```
git add .
```

git commit -m "first commit"

Records the staged changes in Git history with a commit message.

```
git commit -m "first commit"
```

git remote add origin https://github.com/username/repo.git

Connects the local repository to a remote repository.

```
git remote add origin https://github.com/username/repo.git
```

git push -u origin main

Pushes local commits to the remote repository's main branch and sets the upstream relationship.

```
git push -u origin main
```

Git Stages — Easy Concept

Working directory → Staging area → Repository → Remote repository

You make changes in your project, stage them with **git add**, save them in Git history with **git commit**, and send the commits to a remote repository with **git push**.

Complete Git Workflow

Step 1 — Create project folder

```
my-project/
```

Step 2 — Initialize Git

```
git init
```

Step 3 — Add files

```
git add .
```

Step 4 — Commit changes

```
git commit -m "first commit"
```

Step 5 — Connect to GitHub

```
git remote add origin https://github.com/username/repo.git
```

Step 6 — Push code

```
git push -u origin main
```